ScholarGraph API 0.1.0 OAS 3.1

/openapi.json

A research paper intelligence API for paper discovery, citation analysis, and semantic search.

Authorize

default

GET `/health` Health Check

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** | Schema `"string"`	*No links*

Authentication

POST `/auth/register` Register

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "username": "string",
  "email": "user@example.com",
```

```
  "password": "stringst"
}
```

Responses

Code	Description	Links
201	User registered successfully	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "username": "string",
  "email": "user@example.com",
  "created_at": "2026-03-19T01:15:16.253Z"
}
```

400	Email already registered or username already taken	*No links*
422	Invalid registration payload	*No links*

POST /auth/login Login

Authenticate a user and issue JWT tokens.

Expects OAuth2 form data where the `username` field should contain the user's email address and the `password` field contains the plain-text password. Returns both an access token and a refresh token on success.

Parameters Try it out

No parameters

Request body required application/x-www-form-urlencoded

grant_type
`string | (string | null)`
pattern: ^password$

username * required
`string`

password * required
`string($password)`

scope
`string`

client_id
`string | (string | null)`

client_secret

```
string | (string |
null)($password)
```

Responses

Code	Description	Links
200	Authentication succeeded and tokens were issued Media type application/json Controls `Accept` header. **Example Value** | Schema `{ "access_token": "string", "refresh_token": "string", "token_type": "bearer" }`	*No links*
401	Invalid credentials or inactive user	*No links*
422	Invalid login form payload	*No links*

POST `/auth/refresh` Refresh Access Token

Exchange a valid refresh token for a new access token.

Accepts a refresh token in the request body, validates that it is a refresh token belonging to an existing active user, and returns a new access token along with the same refresh token.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "refresh_token": "string"
}
```

Responses

Code	Description	Links
200	Access token refreshed successfully Media type application/json Controls `Accept` header.	*No links*

Code	Description	Links

Example Value | Schema

```
{
  "access_token": "string",
  "refresh_token": "string",
  "token_type": "bearer"
}
```

401	Invalid or expired refresh token	*No links*
422	Invalid refresh token payload	*No links*

GET /auth/me Get Me

Return the currently authenticated active user.

This endpoint is protected by bearer-token authentication and is useful for verifying that JWT-based auth works end-to-end.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Authenticated user returned successfully	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "username": "string",
  "email": "user@example.com",
  "created_at": "2026-03-19T01:15:16.229Z"
}
```

401	Not authenticated	*No links*

GET /auth/api-keys List Api Keys

List API keys for the authenticated user.

Returns metadata only. The raw key and stored key hash are never exposed.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	API keys returned successfully	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
[
  {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "name": "string",
    "scopes": [
      "string"
    ],
    "created_at": "2026-03-19T01:15:16.229Z",
    "last_used_at": "2026-03-19T01:15:16.230Z",
    "is_active": true
  }
]
```

401	Not authenticated	*No links*

POST /auth/api-keys Create Api Key

Create a new API key for the authenticated user.

The raw API key is returned exactly once in this response and is never stored in plain text. Only its SHA-256 hash is stored in the database.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "name": "string",
  "scopes": [
    "string"
  ]
}
```

Responses

Code	Description	Links
201	API key created successfully	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

Code	Description	Links
	`{ "name": "string", "key": "string" }`	
401	Not authenticated	*No links*
422	Invalid API key payload	*No links*

DELETE `/auth/api-keys/{api_key_id}` Revoke Api Key

Revoke an API key owned by the authenticated user.

This marks the key as inactive so it can no longer be used.

Parameters Try it out

Name	Description
api_key_id * required `string($uuid)` *(path)*	api_key_id

Responses

Code	Description	Links
204	API key revoked successfully	*No links*
401	Not authenticated	*No links*
404	API key not found	*No links*
422	Invalid API key ID	*No links*

Crawling

POST `/crawl` Start Crawl

Start a background crawl job for an arXiv topic.

JWT-protected, admin only. Returns 202 immediately while the crawl runs in the background.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "topic": "cs.AI",
  "max_papers": 200
}
```

Responses

Code	Description	Links
202	Crawl job accepted and started in the background Media type application/json Controls Accept header. **Example Value** | Schema	*No links*
401	Not authenticated	*No links*
403	Admin privileges required	*No links*
422	Invalid crawl request payload	*No links*

```
{
  "message": "string",
  "topic": "string",
  "query": "string",
  "max_papers": 0
}
```

POST `/crawl/seed-foundations` Start Seed Foundations

Identify the top-N most-cited papers missing from the corpus and crawl them.

Run this after crawling all topics and before build-graph-all. It finds the foundational papers (BERT, ResNet, Transformers, etc.) that your corpus papers cite heavily but which don't appear in a topic crawl, then fetches and inserts them from arXiv.

Recommended sequence: POST /crawl (repeat for each topic) POST /crawl/seed-foundations POST /crawl/build-graph-all

Parameters Try it out

Name	Description
top_n `integer` *(query)*	*Default value* : 150 150

Responses

Code	Description	Links
202	Foundation seeding job accepted and started in the background Media type application/json Controls Accept header.	*No links*

Code	Description	Links

Example Value | Schema

```
{
  "message": "string",
  "top_n": 0
}
```

401	Not authenticated	*No links*
422	Invalid top_n parameter	*No links*

POST /crawl/build-graph Start Build Graph

Build citation edges for papers in a single topic already in the DB.

Use this for incremental updates after crawling one new topic. For a full corpus build after crawling all topics, use /build-graph-all.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "topic": "cs.AI"
}
```

Responses

Code	Description	Links
202	Graph build job accepted and started in the background	*No links*

Media type
application/json
Controls Accept header.

Example Value | Schema

```
{
  "message": "string",
  "topic": "string"
}
```

401	Not authenticated	*No links*
422	Invalid graph build request payload	*No links*

POST /crawl/build-graph-all Start Build Graph All

Build citation edges across the entire paper corpus in one pass.

Run this once after crawling all topics. Cross-topic citations (e.g. cs.AI paper citing a cs.LG paper) are captured correctly because the full corpus is searched when resolving each reference.

Set force_refresh=true to delete and recreate all existing edges.

Parameters Try it out

Name	Description
force_refresh `boolean` *(query)*	*Default value* : false false

Responses

Code	Description	Links
202	Full corpus graph build accepted and started in the background Media type application/json Controls `Accept` header. **Example Value** | Schema `{ "message": "string", "force_refresh": true }`	*No links*
401	Not authenticated	*No links*
422	Invalid force_refresh parameter	*No links*

Papers

GET `/papers` List Papers

Parameters Try it out

Name	Description
category `string | (string | null)` *(query)*	category
search `string | (string | null)` *(query)*	search
page `integer` *(query)*	*Default value* : 1 1 *minimum: 1*

Name	Description
size integer (query)	*Default value* : 20 20 maximum: 100 minimum: 1

Responses

Code	Description	Links
200	Paginated list of papers returned successfully Media type application/json Controls Accept header. **Example Value** | Schema	*No links*
422	Invalid pagination or filter parameters	*No links*


```
{
  "items": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "arxiv_id": "string",
      "title": "string",
      "abstract": "string",
      "published_date": "2026-03-19"
```


GET `/papers/ranked` Get Ranked Papers Endpoint

Return top papers by PageRank score.

Query params:

- category: optional category/topic filter
- limit: top N papers to return (default 20, max 100)

Parameters Try it out

Name	Description
category `string	(string | null)` *(query)*	category
limit `integer` *(query)*	*Default value* : 20 20 *maximum: 100* *minimum: 1*

Responses

Code	Description	Links
200	Top ranked papers returned successfully Media type **application/json** Controls `Accept` header. **Example Value** | Schema	*No links*
422	Invalid category or limit parameter	*No links*

```
{
  "items": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "arxiv_id": "string",
      "title": "string",
      "abstract": "string",
      "published_date": "2026-03-19"
```

GET /papers/search/semantic Semantic Search Endpoint

Main semantic search endpoint.

Logic:

- sanitize and validate the query through Pydantic
- embed the query
- search papers by vector similarity
- optionally filter by category
- return the top results with similarity_score

Parameters Try it out

Name	Description
q * required `string` *(query)*	q *minLength: 1*
limit `integer` *(query)*	*Default value* : 10 10 *maximum: 100* *minimum: 1*
category `string | (string | null)` *(query)*	category

Responses

Code	Description	Links
200	Semantic search results returned successfully Media type **application/json** Controls `Accept` header. **Example Value** | Schema	*No links*
422	Invalid semantic search query parameters	*No links*

```
{
  "items": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "arxiv_id": "string",
      "title": "string",
      "abstract": "string",
      "published_date": "2026-03-19"
```

GET `/papers/{paper_id}/similar` Get Similar Papers Endpoint

Find papers similar to a given paper using its stored embedding vector.

Excludes the source paper itself.

Parameters

Try it out

Name	Description
paper_id * required `string($uuid)` *(path)*	paper_id
limit `integer` *(query)*	*Default value* : 10 10 *maximum: 100* *minimum: 1*

Responses

Code	Description	Links
200	Similar papers returned successfully Media type application/json Controls Accept header. **Example Value** | Schema	*No links*
400	Paper exists but does not have an embedding yet	*No links*
404	Paper not found	*No links*
422	Invalid paper ID or limit parameter	*No links*

```
{
  "items": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "arxiv_id": "string",
      "title": "string",
      "abstract": "string",
      "published_date": "2026-03-19"
```

GET `/papers/{paper_id}` Get Paper By Id

Parameters

Try it out

Name	Description
paper_id * required `string($uuid)` *(path)*	paper_id

Responses

Code	Description	Links
200	Paper returned successfully Media type **application/json** Controls `Accept` header. **Example Value** | Schema	*No links*

```
{
  "_links": {
    "additionalProp1": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    },
    "additionalProp2": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    },
    "additionalProp3": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    }
  },
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "arxiv_id": "string",
  "title": "string",
  "abstract": "string",
  "published_date": "2026-03-19",
  "updated_date": "2026-03-19T01:15:16.239Z",
  "primary_category": "string"
```

404	Paper not found	*No links*
422	Invalid paper ID	*No links*

GET `/papers/{paper_id}/citations` Get Paper Citations Endpoint

Return both citation directions for a paper:

- cited_by: papers that cite this paper
- references: papers this paper cites

Parameters Try it out

Name	Description
paper_id * required `string($uuid)` *(path)*	paper_id

Name	Description
page **integer** *(query)*	*Default value* : 1 1 *minimum: 1*
size **integer** *(query)*	*Default value* : 20 20 *maximum: 100* *minimum: 1*

Responses

Code	Description	Links
200	Citation relationships returned successfully Media type **application/json** Controls `Accept` header. **Example Value** | Schema	*No links*
404	Paper not found	*No links*
422	Invalid paper ID or pagination parameters	*No links*


```
{
  "items": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "arxiv_id": "string",
      "title": "string",
      "abstract": "string",
      "published_date": "2026-03-19"
```


GET `/papers/{paper_id}/authors` Get Paper Authors Endpoint

Return authors for a paper, including their position on the paper.

Parameters Try it out

Name	Description
paper_id * required `string($uuid)` *(path)*	paper_id
page `integer` *(query)*	*Default value* : 1 1 *minimum: 1*
size `integer` *(query)*	*Default value* : 20 20 *maximum: 100* *minimum: 1*

Responses

Code	Description	Links
200	Paper authors returned successfully Media type **application/json** Controls Accept header. **Example Value** | Schema	*No links*
404	Paper not found	*No links*
422	Invalid paper ID or pagination parameters	*No links*

```
{
  "items": [
    {
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "name": "string",
      "position": 0
    }
  ],
  "total": 0,
  "page": 0,
  "size": 0
}
```

Annotations

POST `/papers/{paper_id}/annotations` Create Annotation

Create an annotation for a paper.

The authenticated user becomes the annotation owner.

Parameters Try it out

Name	Description
paper_id * required `string($uuid)`	paper_id

Name	Description
(path)	

Request body required

application/json

Example Value | Schema

```
{
  "title": "string",
  "body": "string",
  "tags": [
    "string"
  ]
}
```

Responses

Code	Description	Links
201	Annotation created successfully Media type application/json Controls Accept header. **Example Value** | Schema	*No links*
401	Not authenticated	*No links*
404	Paper not found	*No links*
422	Invalid annotation payload or paper ID	*No links*

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "paper_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "title": "string",
  "body": "string",
  "tags": [
    "string"
  ],
  "created_at": "2026-03-19T01:15:16.242Z",
  "updated_at": "2026-03-19T01:15:16.242Z"
}
```

GET /papers/{paper_id}/annotations List Annotations For Paper

Return all annotations for a paper.

This endpoint is public.

Parameters Try it out

Name	Description
paper_id * required `string($uuid)` *(path)*	paper_id

Responses

Code	Description	Links
200	Annotations returned successfully	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
[
  {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "paper_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "title": "string",
    "body": "string",
    "tags": [
      "string"
    ],
    "created_at": "2026-03-19T01:15:16.243Z",
    "updated_at": "2026-03-19T01:15:16.243Z"
  }
]
```

404	Paper not found	*No links*
422	Invalid paper ID	*No links*

PUT /annotations/{annotation_id} Update Annotation

Update an annotation.

Only the annotation owner may update it.

Parameters

Try it out

Name	Description
annotation_id * required `string($uuid)` *(path)*	annotation_id

Request body required

application/json

Example Value | Schema

```
{
  "title": "string",
  "body": "string",
  "tags": [
    "string"
  ]
}
```

Responses

Code	Description	Links
200	Annotation updated successfully	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "paper_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "title": "string",
  "body": "string",
  "tags": [
    "string"
  ],
  "created_at": "2026-03-19T01:15:16.244Z",
  "updated_at": "2026-03-19T01:15:16.244Z"
}
```

401	Not authenticated	*No links*
403	You can only update your own annotations	*No links*
404	Annotation not found	*No links*
422	Invalid annotation payload or annotation ID	*No links*

DELETE /annotations/{annotation_id} Delete Annotation

Delete an annotation.

Only the annotation owner may delete it.

Parameters Try it out

Name	Description
annotation_id * required `string($uuid)` *(path)*	annotation_id

Responses

Code	Description	Links
204	Annotation deleted successfully	*No links*
401	Not authenticated	*No links*
403	You can only delete your own annotations	*No links*
404	Annotation not found	*No links*
422	Invalid annotation ID	*No links*

Authors

GET /authors List Authors

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Authors returned successfully Media type: application/json Controls Accept header.	*No links*

Example Value | Schema

```
{
  "items": [
    {
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "name": "string",
      "paper_count": 0,
      "avg_pagerank_score": 0
    }
  ],
  "total": 0
}
```

GET /authors/{author_id} Get Author By Id

Parameters Try it out

Name	Description
author_id * required `string($uuid)` *(path)*	author_id

Responses

Code	Description	Links
200	Author returned successfully Media type: application/json Controls Accept header.	*No links*

Example Value | Schema

```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
```

Code		Links


```
  "name": "string",
  "paper_count": 0,
  "avg_pagerank_score": 0,
  "papers": [
    {
      "_links": {
        "additionalProp1": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp2": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        },
        "additionalProp3": {
          "additionalProp1": "string",
          "additionalProp2": "string",
          "additionalProp3": "string"
        }
      },
```


404	Author not found	*No links*
422	Invalid author ID	*No links*

GET /authors/{author_id}/impact Get Author Impact

Return author citation analytics:

- total papers
- total citations received
- top 5 papers by pagerank_score
- average pagerank across all papers

Parameters

Try it out

Name	Description
author_id * required `string($uuid)` *(path)*	author_id

Responses

Code	Description	Links
200	Author impact analytics returned successfully	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema


```
{
  "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "name": "string",
  "total_papers": 0,
  "total_citations_received": 0,
  "avg_pagerank_score": 0,
  "top_papers": [
```

Code	Description	Links

```
{
  "_links": {
    "additionalProp1": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    },
    "additionalProp2": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    },
    "additionalProp3": {
      "additionalProp1": "string",
      "additionalProp2": "string",
      "additionalProp3": "string"
    }
```

404	Author not found	*No links*
422	Invalid author ID	*No links*

Analytics

POST `/analytics/pagerank` Start Pagerank

Trigger a background PageRank computation.

Admin-only, JWT-protected. Returns immediately with 202 Accepted while the computation continues in the background. Progress is logged to the console.

Parameters Try it out

No parameters

Responses

Code	Description	Links
202	PageRank job accepted and started in the background	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "message": "string"
}
```

401	Not authenticated	*No links*
403	Admin privileges required	*No links*
422	Invalid request	*No links*

GET `/analytics/topics` Get Topic Analytics

Return topic/category analytics:

- paper count per primary_category
- average pagerank_score per primary_category

Parameters Try it out

Name	Description
limit `integer` *(query)*	*Default value* : 50 50 *maximum: 200* *minimum: 1*

Responses

Code	Description	Links
200	Topic analytics returned successfully Media type application/json Controls `Accept` header. **Example Value** | Schema	*No links*
422	Invalid limit parameter	*No links*

```
{
  "items": [
    {
      "category": "string",
      "paper_count": 0,
      "avg_pagerank_score": 0
    }
  ],
  "total": 0
}
```

GET `/analytics/trend` Get Publication Trend

Return publication count over time, optionally filtered by topic.

Uses SQL DATE_TRUNC to bucket papers by month or year based on published_date.

Parameters Try it out

Name	Description
topic `string | (string | null)` *(query)*	Filter by primary_category, e.g. cs.AI topic

Name	Description
granularity `string` *(query)*	Time bucket size *Available values* : month, year *Default value* : month month

Responses

Code	Description	Links
200	Publication trend returned successfully Media type application/json Controls Accept header. **Example Value** | Schema	*No links*
422	Invalid topic or granularity parameter	*No links*

```
{
  "topic": "string",
  "granularity": "month",
  "items": [
    {
      "period": "2026-03-19T01:15:16.249Z",
      "paper_count": 0
    }
  ],
  "total_points": 0
}
```

POST /analytics/embed-papers Start Embed Papers

Trigger background embedding for all papers whose abstract_embedding is NULL.

Parameters Try it out

No parameters

Responses

Code	Description	Links
202	Paper embedding job accepted and started in the background Media type application/json Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "message": "string"
```

Code	Description	Links
	`}`	
401	Not authenticated	*No links*
403	Admin privileges required	*No links*
422	Invalid request	*No links*

Schemas

APIKeyCreate > Expand all `object`

APIKeyListItem > Expand all `object`

APIKeyResponse > Expand all `object`

AnnotationCreate > Expand all `object`

AnnotationResponse > Expand all `object`

AnnotationUpdate > Expand all `object`

AuthorDetailResponse > Expand all `object`

AuthorImpactResponse > Expand all `object`

AuthorListItem > Expand all `object`

AuthorListResponse > Expand all `object`

Body_login_auth_login_post > Expand all `object`

BuildGraphAcceptedResponse > Expand all `object`

BuildGraphAllAcceptedResponse > Expand all `object`

BuildGraphRequest > Expand all `object`

CitationPaperList > Expand all `object`

CitationPaperResponse > Expand all `object`

CrawlAcceptedResponse > Expand all `object`

CrawlRequest > Expand all `object`

EmbedPapersAcceptedResponse > Expand all `object`

PageRankAcceptedResponse > Expand all `object`

PaperAuthorList > Expand all `object`

PaperAuthorResponse > Expand all `object`

PaperList > Expand all `object`

PaperResponse > Expand all `object`

RankedPaperList > Expand all `object`

RankedPaperResponse > Expand all `object`

RefreshTokenRequest > Expand all `object`

SeedFoundationsAcceptedResponse > Expand all `object`

SemanticSearchPaperList > Expand all `object`

SemanticSearchPaperResponse > Expand all `object`

Token > Expand all `object`

TopicAnalyticsItem > Expand all `object`

TopicAnalyticsResponse > Expand all `object`

TrendPoint > Expand all `object`

TrendResponse > Expand all `object`

UserCreate > Expand all `object`

UserResponse > Expand all `object`